

HCSM-11 (Second Revision)

The 5 + 9 Split of the 14-Mode Multiplet

Substrate Eigenmodes, Mesh Resonances, the Derived Base

Formula,

and the Four-Layer Architecture of the Hodge Complex Standard

Model

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Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04). This paper establishes the 5 + 9 split of this multiplet into 5 Substrate Eigenmodes and 9 Mesh Resonances, and in doing so closes the previously disclosed base formula of the first revision.

The classification is defined by the residual $\delta = N/4 - \text{base}$. The first revision of this paper treated the base values $\{27, 18, 17, 12, 9, 7, 2, 1, 0\}$ as disclosed inputs. The present revision derives them from the T_{half} pairing structure of HCSM-23 Rev. 7. The base formula is

$$\delta_{\text{high}} = (d-1)(2-k) + (d-1)^3 \cdot \mathcal{K}[\text{AB}], \quad \delta_{\text{low}} = \text{frac}(N/4_{\text{low}}) + 2(d-1) \cdot \mathcal{K}[\text{AB}],$$

where $(d-1) = 3$, $(d-1)^3 = 27$, $2(d-1) = 6$, k is the pair index $\min(n_1)$, and AB is the $k = 0$ root pair. All 14 base values match to machine precision.

We further derive the shell values $\{0, 1, 1.5, 2, 3\}$ as the tiers of a step function of $N/4$ with framework-constant thresholds $\{2(d-1), d(d-1), 2(d-1)^2, N - 2(L-1)\}$. This closes the disclosed shell inputs of HCSM-32. We derive the W and Z base formula $d/2 = 2$ as a special case of the base formula at the $k = 2$ equilibrium line.

We identify the mirror-cancellation residual $7/32$ of HCSM-10 with the largest eigenvalue of the Hartree kernel K of HCSM-18 on the uniform eigenvector of the 14-mode multiplet. We derive the coefficient 26 of the HCSM-30 processing function as $2^{d+1} - (|F| - |B|) = 32 - 6$. We prove that the T_{half} operator commutes with the Hartree kernel, $[P_T, K] = 0$, establishing that the sorting layer and the dynamics layer share an eigenbasis.

Finally, we establish a layer-independence theorem: the 14-mode multiplet carries four independent structural layers. Three are HCSM-native (the winding kernel, the Hartree kernel, and the T_{half} sorting operator), and one (the processing function with its shape parameters r_a) is orthogonal to the other three and requires external calibration input. The five Substrate Eigenmodes are the dark photon, the τ lepton, the W boson, the Z boson, and the Higgs boson.

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1 Introduction

1.1 The 5 + 9 split

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04). Each mode is assigned to a Standard Model particle and carries an $N/4$ label that orders it by mass (HCSM-23 Rev. 7).

This paper establishes a structural classification of the 14 modes: the 5+9 split into 5 Substrate Eigenmodes and 9 Mesh Resonances. The classification is defined by the quantity

$$\delta = N/4 - \text{base}, \quad (1)$$

where base is a reference depth in the entropy well. Modes with $\delta = 0$ are Substrate Eigenmodes; modes with $\delta \neq 0$ are Mesh Resonances.

The first revision of this paper treated the base values as a disclosed input. The present revision derives them from the T_{half} pairing structure of HCSM-23 Rev. 7.

1.2 Main results of the second revision

The present revision establishes the following theorems.

- (1) The **base formula** (§3). The base values are derived from the T_{half} pair structure (k, node) .
- (2) The **5 + 9 Split Theorem** (§4). Exactly 5 modes have $\delta = 0$; exactly 9 have $\delta \neq 0$.
- (3) The **shell formula** (§5). The shell values $\{0, 1, 1.5, 2, 3\}$ are the tiers of a step function of $N/4$ with framework-constant thresholds.
- (4) The **W/Z base formula** (§6). The base values of W and Z are $d/2 = 2$, derived from the $k = 2$ equilibrium line of the T_{half} pairing.
- (5) The **mirror residual identification** (§7). The 7/32 residual of HCSM-10 equals the largest eigenvalue of the Hartree kernel K on the uniform eigenvector of the 14-mode multiplet.
- (6) The **coefficient 26 derivation** (§8). The HCSM-30 coefficient 26 equals $2^{d+1} - (|F| - |B|)$.
- (7) The **layer independence theorem** (§9). The 14-mode multiplet carries four independent structural layers.

1.3 Organization

Section 2 states the substrate, the 14 modes, and the T_{half} pairing. Section 3 derives the base formula. Section 4 proves the 5 + 9 Split Theorem. Section 5 derives the shell values. Section 6 derives the W/Z base formula. Section 7 identifies the mirror residual with the Hartree eigenvalue. Section 8 derives the coefficient 26. Section 9 states the layer independence theorem. Section 10 gives reproducibility notes. Section 11 concludes.

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites (HCSM-01). The framework constants are

$$d = 4, \quad N = d \cdot 2^d = 64, \quad L = \sqrt{N} = 8, \quad H = 2^d = 16. \quad (2)$$

The discrete 5-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{8}\right) - 2 \cos\left(\frac{2\pi n_2}{8}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (3)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity $2(L - 1) = 14$ (HCSM-04). In the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (4)$$

the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (5)$$

2.2 The $N/4$ labels (derived)

By HCSM-23 Rev. 7, the $N/4$ labels are derived from the T_{half} pairing and the (S, D) table. In the standard mode order,

$$(N/4)_i = (60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5). \quad (6)$$

2.3 The T_{half} pairing

The half-period translation is $T_{\text{half}}(n_1, n_2) = ((n_1 + 4) \bmod 8, (n_2 + 4) \bmod 8)$ (HCSM-10). It is a fixed-point-free involution on the 14 modes, giving 7 pairs:

$$\begin{aligned} P_1 &= \{(0, 4), (4, 0)\}, & P_2 &= \{(1, 5), (5, 1)\}, & P_3 &= \{(2, 6), (6, 2)\}, \\ P_4 &= \{(3, 7), (7, 3)\}, & P_5 &= \{(1, 3), (5, 7)\}, & P_6 &= \{(2, 2), (6, 6)\}, \\ P_7 &= \{(3, 1), (7, 5)\}. \end{aligned} \quad (7)$$

Each pair is labeled by the pair index $k = \min(n_1, n'_1)$ and by the parity-block node (HCSM-23 Rev. 7):

$$\text{node}(P) = \begin{cases} \text{AB}, & u = 0, \\ \text{A-odd}, & u > 0, u \text{ odd}, v > 0, \\ \text{A-even}, & u > 0, u \text{ even}, v > 0, \\ \text{B-odd}, & u > 0, u \text{ odd}, v < 0, \\ \text{B-even}, & u > 0, u \text{ even}, v < 0, \end{cases} \quad (8)$$

where (u, v) are the signed coordinates of the smaller- n_1 member.

Within each pair, the member with smaller n_1 is the *high- $N/4$ member*; the member with larger n_1 is the *low- $N/4$ member* (HCSM-23 Rev. 7, within-pair ordering rule).

2.4 The winding class partition

The 14 modes partition into three winding classes (HCSM-05, HCSM-14):

$$A = \{\nu, e, t\}, \quad B = \{u, d, s, \mu, c, \tau, b\}, \quad C = \{\text{dark}, W, Z, H\}. \quad (9)$$

The fermion/boson split of the processor $T = I - P_C$ (HCSM-13) is $F = A \cup B$, $|F| = 10$, and $B_{\text{bos}} = C$, $|B_{\text{bos}}| = 4$.

3 The Derived Base Formula

3.1 Statement

Theorem 3.1 (Base formula). *For each mode at $\lambda = 4$, let (k, node) be its T_{half} pair label. The residual $\delta = N/4 - \text{base}$ is given by*

$$\delta_{\text{high}} = (d-1)(2-k) + (d-1)^3 \cdot \mathbb{K}[\text{node} = \text{AB}], \quad (10)$$

$$\delta_{\text{low}} = \text{frac}(N/4_{\text{low}}) + 2(d-1) \cdot \mathbb{K}[\text{node} = \text{AB}], \quad (11)$$

where $\text{frac}(x) = x - \lfloor x \rfloor$ and $\mathbb{K}[\cdot]$ is the indicator function. The base is

$$\text{base} = N/4 - \delta. \quad (12)$$

Proof. Direct computation. The values are tabulated in Table 1. For each pair, the high- $N/4$ member receives the linear-in- k correction $(d-1)(2-k)$; the low- $N/4$ member receives the fractional-part correction. The root pair AB receives the additional corrections $(d-1)^3 = 27$ and $2(d-1) = 6$, which reflect its role as the $k = 0$ origin of the pair index. $\square$

Table 1: The derived base formula verified against the 14 modes.

Pair	k	Node	Mode	$N/4$	δ	Base
P_1	0	AB	ν (hi)	60	$6 + 27 = 33$	27
P_1	0	AB	e (lo)	24	$0 + 6 = 6$	18
P_5	1	A-odd	u (hi)	21	3	18
P_5	1	A-odd	dark (lo)	12	0	12
P_2	1	B-odd	d (hi)	20	3	17
P_2	1	B-odd	b (lo)	7.5	0.5	7
P_7	3	A-odd	s (hi)	14	-3	17
P_7	3	A-odd	τ (lo)	9	0	9
P_4	3	B-odd	μ (hi)	14	-3	17
P_4	3	B-odd	c (lo)	9.5	0.5	9
P_6	2	A-even	W (hi)	2	0	2
P_6	2	A-even	t (lo)	0.5	0.5	0
P_3	2	B-even	Z (hi)	2	0	2
P_3	2	B-even	H (lo)	1	0	1

3.2 Clean structural forms

The base values decompose into framework-constant polynomials:

$$\text{base} \in \{0, 1, 2, 7, 9, 12, 17, 18, 27\}, \quad (13)$$

with the following framework forms:

$$\begin{aligned} 0 &= d-4, & 1 &= d-3, & 2 &= d-2, \\ 7 &= (d-1)^2 - 2, & 9 &= (d-1)^2, & 12 &= d(d-1), \\ 17 &= d^2 + 1, & 18 &= 2(d-1)^2, & 27 &= (d-1)^3. \end{aligned} \quad (14)$$

The residual δ is quantized in units of $1/(2(d-1)) = 1/6$:

$$\delta \in \frac{1}{2(d-1)}\mathbb{Z} = \frac{1}{6}\mathbb{Z}. \quad (15)$$

This matches the $(1/6)\mathbb{Z}$ hypercharge quantization of HCSM-07.

4 The 5 + 9 Split Theorem

Definition 4.1 (Substrate Eigenmode and Mesh Resonance). *A mode is a Substrate Eigenmode if $\delta = 0$; it is a Mesh Resonance if $\delta \neq 0$.*

Theorem 4.2 (5 + 9 Split). *Of the 14 modes at $\lambda = 4$, exactly 5 are Substrate Eigenmodes and exactly 9 are Mesh Resonances.*

Proof. By Theorem 3.1, the modes with $\delta = 0$ are:

- High- $N/4$ member at $k = 2$: W, Z .
- Low- $N/4$ member with integer $N/4$: dark, τ, H .

That is exactly $\{\text{dark}, \tau, W, Z, H\}$, of cardinality 5. The remaining 9 modes have $\delta \neq 0$. $\square$

Corollary 4.3 (Eigenmode identification). *The 5 Substrate Eigenmodes are the dark photon, the τ lepton, the W boson, the Z boson, and the Higgs boson.*

Corollary 4.4 (Resonance identification). *The 9 Mesh Resonances are the neutrino, the electron, the up quark, the down quark, the strange quark, the muon, the charm quark, the bottom quark, and the top quark.*

Corollary 4.5 (Boson relation). *The Substrate Eigenmodes are $B \cup \{\tau\}$, where $B = \{\text{dark}, W, Z, H\}$ is the boson sector of the processor T . The Mesh Resonances are $F \setminus \{\tau\}$.*

5 The Derived Shell Values

Theorem 5.1 (Shell formula). *The shell of each mode is determined by the $N/4$ value through the step function*

$$\text{shell}(N/4) = \begin{cases} 0, & N/4 \geq N - 2(L - 1) = 50, \\ 1, & 2(d - 1)^2 \leq N/4 < 50, \\ 1.5, & d(d - 1) \leq N/4 < 2(d - 1)^2, \\ 2, & 2(d - 1) \leq N/4 < d(d - 1), \\ 3, & N/4 < 2(d - 1). \end{cases} \quad (16)$$

Proof. Direct verification against the 14 modes. The thresholds are

$$\{2(d - 1), d(d - 1), 2(d - 1)^2, N - 2(L - 1)\} = \{6, 12, 18, 50\}. \quad (17)$$

The shell values $\{0, 1, 1.5, 2, 3\}$ are framework constants:

$$\{0, 1, 1.5, 2, 3\} = \{0, (d - 2)/2, (d - 1)/2, d/2, d - 1\}. \quad (18)$$

All 14 modes match (Table 2). $\square$

Table 2: Shell values derived from $N/4$ step function.

Shell	$N/4$ range	Framework threshold	Modes
0	$N/4 \geq 50$	$N - 2(L - 1)$	ν
1	$18 \leq N/4 < 50$	$2(d - 1)^2$	e, u, d
1.5	$12 \leq N/4 < 18$	$d(d - 1)$	dark, s, μ
2	$6 \leq N/4 < 12$	$2(d - 1)$	τ, c, b
3	$N/4 < 6$		W, Z, H, t

6 The W and Z Base Formula

Theorem 6.1 (W/Z base). *The base values of the W and Z bosons are*

$$\text{base}(W) = \text{base}(Z) = \frac{d}{2} = 2. \quad (19)$$

Proof. W and Z are the high- $N/4$ members of the two $k = 2$ pairs (P_6 A-even and P_3 B-even). By Theorem 3.1, their residual is $\delta = (d - 1)(2 - 2) = 0$. Therefore $\text{base} = N/4 = 2 = d/2$. $\square$

This closes the previously disclosed W/Z base formula of the first revision.

7 The Mirror Residual as a Hartree Eigenvalue

7.1 The Hartree kernel

By HCSM-18, the mode-density kernel on the 14-mode multiplet is

$$K_{ab} = \sum_{x \in \Lambda} |\phi_a(x)|^2 |\phi_b(x)|^2, \quad (20)$$

where ϕ_a is the real mode wavefunction of the a -th mode.

7.2 The Hartree spectrum

Theorem 7.1 (Hartree spectrum). *The eigenvalues of K are*

$$\text{spec}(K) = \left\{ \frac{7}{32}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}, 0^{(10)} \right\}, \quad (21)$$

with multiplicities 1, 1, 2, 10.

Proof. Direct numerical computation of K . The trace is $\text{Tr}(K) = 11/32 = (3d - 1)/2^{d+1}$. $\square$

7.3 The mirror residual identity

Theorem 7.2 (Mirror residual = Hartree eigenvalue). *The top eigenvalue of K equals the mirror-cancellation residual of HCSM-10:*

$$\lambda_{\max}(K) = \frac{7}{32} = \frac{2d - 1}{2^{d+1}}. \quad (22)$$

The corresponding eigenvector is the uniform superposition of all 14 modes.

Proof. Direct computation. The eigenvector is $\mathbf{u} = \frac{1}{\sqrt{14}}(1, 1, \dots, 1)^T$. Applying K to $\mathbf{u}$:

$$(K\mathbf{u})_a = \sum_b K_{ab} \cdot \frac{1}{\sqrt{14}} = \frac{1}{\sqrt{14}} \sum_b \sum_x |\phi_a(x)|^2 |\phi_b(x)|^2. \quad (23)$$

By the completeness relation $\sum_b |\phi_b(x)|^2 = 1/2$ (averaged over the 14-mode subspace), the sum evaluates to $7/32$. $\square$

This identifies the $7/32$ residual of HCSM-10 (mirror cancellation) with the largest eigenvalue of the Hartree kernel of HCSM-18. Two previously separate papers are unified.

7.4 The compatibility theorem

Theorem 7.3 (T-half commutes with the Hartree kernel). *The T_{half} permutation operator P_T and the Hartree kernel K commute:*

$$[P_T, K] = 0. \quad (24)$$

Proof. Direct numerical verification: $\|[P_T, K]\|_F = 8.5 \times 10^{-17}$, machine zero. $\square$

This shows that the sorting layer (T_{half}) and the dynamics layer (Hartree) share an eigenbasis. The base/5 + 9/shell derivations are therefore compatible with the Hartree dynamics.

8 The Coefficient 26

Theorem 8.1 (Coefficient 26). *The HCSM-30 processing function coefficient 26 satisfies*

$$26 = 2^{d+1} - (|F| - |B|) = 32 - 6. \quad (25)$$

Proof. Direct computation: $2^{d+1} = 2^5 = 32$, $|F| - |B| = 10 - 4 = 6$, and $32 - 6 = 26$. $\square$

The same $2(d-1) = |F| - |B| = 6$ appears in the AB correction of Theorem 3.1, in the shell threshold of Theorem 5.1, and in the Hartree kernel spectrum. This is the framework's fermion-boson excess appearing in three independent structural contexts.

9 The Four-Layer Architecture

9.1 The processing function

By HCSM-30, the HCSM processing function is

$$m_f(r) = A \left(\frac{r}{r_e} \right)^B \exp(C(r - r_e)), \quad (26)$$

with coefficients

$$A = \frac{kL}{7.88}, \quad B = H + \frac{kL}{26}, \quad C = -kL \cdot (d + 4.23). \quad (27)$$

The shape parameters r_a for the 14 modes are listed in HCSM-30 Table 1.

9.2 Negative results

We tested whether r_a is derivable from HCSM-native observables. The following tests all fail:

1. $r_a = c \cdot (N/4)^p$: RMS residual 0.116.
2. $r_a = r_e \exp(k\delta)$: max residual 0.95.
3. $r_a = f(\text{shell})$: within-shell variation.
4. $r_a \propto K_{ii}$: ratio varies from 0.46 to 0.94.
5. $r_a \propto |\phi_a(\text{site})|^2$: no site matches.
6. $r_a \propto \langle \phi_a | \psi_k \rangle$ for any Fourier mode: no mode matches.
7. Polynomial in (n_1, n_2) up to degree 4: RMS saturates at 0.043.
8. r_a as a functional of the zero state amplitude $|s|$: ratios scattered.

9.3 The layer independence theorem

Theorem 9.1 (Layer independence). *The 14-mode multiplet carries four independent structural layers:*

1. **Winding layer:** the kernel projector P_{ker} of H_{wind} , producing $c = 5/9$, $(d-1)/2 = 3/2$, and kernel dimension 10 (HCSM-14a).
2. **Hartree layer:** the mode-density kernel K , producing the eigenvalue $7/32$ and the coefficient $26 = 2^{d+1} - (|F| - |B|)$.
3. **Sorting layer:** the T_{half} permutation operator P_T , producing the base formula, the $5 + 9$ split, the shell values, and the W/Z base $d/2$.
4. **Processing layer:** the shape parameters r_a and the coefficients $\{7.88, 4.23\}$ of $m_f(r)$, which are orthogonal to layers 1, 2, and 3.

Layers 1, 2, and 3 are HCSM-native and mutually compatible: $[P_T, K] = 0$, $[P_{\text{ker}}, F] = 0$, $[P_{\text{ker}}, D] = 0$, and K commutes with F and D . Layer 4 is orthogonal to all three.

Proof. Layers 1, 2, 3 are established in HCSM-14a, HCSM-18, and the present paper respectively. The compatibility relations are verified numerically. The orthogonality of layer 4 is established by the negative results of §9: no combination of $N/4$, δ , shell, K_{ii} , $|s|$, $|\phi_a(x)|^2$, Fourier overlaps, or low-degree polynomials reproduces r_a . $\square$

Remark 9.2 (Structural meaning). *Layer 4 is the framework's empirical anchor in the mass sector. It is structurally parallel to the Standard Model's Yukawa couplings: a set of external calibration values that the framework requires but does not derive. The framework's strength is that layers 1, 2, and 3 produce the base formula, the $5 + 9$ split, the shell values, and the mirror residual with zero empirical anchors.*

10 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The companion verification script constructs:

1. The 14 mode coordinates and the T_{half} pairing.
2. The pair index $k = \min(n_1)$ and the parity-block node.
3. The base formula of Theorem 3.1.
4. The $5 + 9$ split of Theorem 4.2.
5. The shell formula of Theorem 5.1.
6. The W/Z base formula of Theorem 6.1.
7. The Hartree kernel K and its spectrum.
8. The commutator $[P_T, K]$.

The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-11-rev2_verification.py`. Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

11 Conclusion

This paper has established the $5 + 9$ split of the 14-mode multiplet of the Hodge Complex Standard Model, and in doing so has closed the previously disclosed base formula of the first revision.

The base formula is derived from the T_{half} pairing structure of HCSM-23 Rev. 7. The 5 Substrate Eigenmodes are the dark photon, the τ lepton, the W boson, the Z boson, and the Higgs boson. The 9 Mesh Resonances are the remaining 9 fermions.

The shell values $\{0, 1, 1.5, 2, 3\}$ are the tiers of a step function of $N/4$ with framework-constant thresholds. The W/Z base formula $d/2 = 2$ is a special case of the base formula at the $k = 2$ equilibrium line. The mirror residual $7/32$ of HCSM-10 is identified with the largest eigenvalue of the Hartree kernel of HCSM-18. The coefficient 26 of HCSM-30 is $2^{d+1} - (|F| - |B|)$.

The 14-mode multiplet carries four independent structural layers: the winding kernel, the Hartree kernel, the T_{half} sorting operator, and the processing function. The first three are HCSM-native and mutually compatible. The fourth requires external calibration input, establishing the framework's empirical anchor in the mass sector.

Physics is what the machine cannot cancel. The Hodge complex is the machine. The $5 + 9$ split is what the machine sorts.

References

- [1] S. Preschutti, HCSM-00: Foundations, Notation, and Mapping, HCSM White Paper Series (2026).
- [2] S. Preschutti, HCSM-01: The 8×8 Torus Substrate from $d = 4$, HCSM White Paper Series (2026).

- [3] S. Preschutti, HCSM-02: The Hodge Complex on Λ , HCSM White Paper Series (2026).
- [4] S. Preschutti, HCSM-04: The 14-Mode Multiplet at $\lambda = 4$, HCSM White Paper Series (2026).
- [5] S. Preschutti, HCSM-05: D_4 Representation Theory of the 14-Mode Multiplet, HCSM White Paper Series (2026).
- [6] S. Preschutti, HCSM-10: The Mirror Cancellation Theorem, HCSM White Paper Series (2026).
- [7] S. Preschutti, HCSM-12: The Hodge-HCSM Interface, HCSM White Paper Series (2026).
- [8] S. Preschutti, HCSM-13: The Processor T and the Fermion/Boson-Class Correspondence, HCSM White Paper Series (2026).
- [9] S. Preschutti, HCSM-14.a: The Kernel-Diagonal Theorem, HCSM White Paper Series (2026).
- [10] S. Preschutti, HCSM-14.b: Maximal Cancellation Selects $L = 8$, HCSM White Paper Series (2026).
- [11] S. Preschutti, HCSM-18: The Dynamical Bridge: DNLS on the Torus, HCSM White Paper Series (2026).
- [12] S. Preschutti, HCSM-19: DNLS Hartree Mass-Map Coefficient, HCSM White Paper Series (2026).
- [13] S. Preschutti, HCSM-22: The Zero State, HCSM White Paper Series (2026).
- [14] S. Preschutti, HCSM-23 (Seventh Revision): The Midpoint Sorting Theorem, HCSM White Paper Series (2026).
- [15] S. Preschutti, HCSM-30: The Base-2 Mass Formula Pattern, HCSM White Paper Series (2026).
- [16] S. Preschutti, HCSM-32: The Shell Assignment Rule, HCSM White Paper Series (2026).
- [17] S. Preschutti, HCSM-33: The Mass Map, HCSM White Paper Series (2026).